L-P-385C
October 7, 1974
SUPERSEDING
Fed. Spec. L-P-385B
October 15, 1968

FEDERAL SPECIFICATION

PLASTIC MOLDING AND EXTRUSION MATERIAL, POLYCHLOROTRIFLUOROETHYLENE

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers the requirements for polychlorotrifluoroethylene plastics suitable for injection molding, compression molding, and extrusion.

1.2 Classification.

1.2.1 Types, classes, and grades (as applicable). The polychlorotrifluoroethylene plastic material shall be of the following types, classes, and grades, as specified (see 6.2).

Type I - Homopolymer.

Class 2 - Zero strength time range 176 to 300 seconds.
Class 3 - Zero strength time range 301 to 750 seconds.

Type II - Modified copolymer.

Class 1 - Zero strength time range 115 to 175 seconds.
Class 2 - Zero strength time range 176 to 300 seconds.
Class 3 - Zero strength time range 301 to 750 seconds.
Class 4 - Zero strength time range 751 to 2500 seconds.

FSC 9330

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Type III - Copolymer.

Class 2 - Zero strength time range 176 to 300 seconds

Grade A - Higher melting point than Grade B.
Grade B - Lower melting point than Grade A.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

Federal Specifications:

PPP-D-723 - Drums, Fiber
PPP-D-729 - Drums, Shipping and Storage, Steel, 55-Gallon

Federal Standards:

Fed. Std. No. 123 - Marking for Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, D. C. 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge from the Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standards:

MIL-STD-105 - Sampling procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials (ASTM) Standards:

D 150	- A. C. Loss Characteristics and Dielectric Constant (Permittivity) of Solid Electrical Insulating Materials.
D 618	- Conditioning Plastics and Electrical Insulating Materials for Testing.
D 621	- Deformation of Plastics Under Load.
D 792	- Specific Gravity and Density of Plastics by Displacement.
D 1430	- Polychlorotrifluoroethylene (PCTFE) Plastics.
D 2117	- Melting Point of Semicrystalline Polymers.

(Application for copies should be addredded to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pa. 19103).

National Motor Freight Traffic Association, Inc., Agent:
National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Associations, Inc., Tariff Order Section, 1616 P Street, N.W., Washington, DC 20036.)

Uniform Classification Committee, Agent:
Uniform Freight Classification

(Applications for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, Illinois 60606.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Materials. The material shall consist of polychlorotrifluoroethylene plastics with a minimum of 90 percent chlorotrifluoroethylene. The remaining maximum of 10 percent may include chemical modifications such as co-monomers, but not colorants, fillers, plasticizers, or mechanical blends with other resins. Material reprocessed from fabricated articles or waste shall not be included.

3.2 Form. The material shall be supplied as pelletized resin, granular resin, or powder resin, as specified by the procuring agency (see 6.2 and 6.3) When a particular particle size is specified, the limits and method of testing shall be as specified by the procuring agency (see 6.2).

3.3 Property values. When tested as specified in the applicable procedure of 4.3, the material shall conform to the property values shown in table 1.

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TABLE I. Property values

Property	Type I		Type II				Type III	
							Class 2	
	Class 2	Class 3	Class 1	Class 2	Class 3	Class 4	Grade A	Grade B
Specific gravity range	2.10 to 2.12	2.10 to 2.12	2.10 to 2.12	2.10 to 2.12	2.10 to 2.12	2.10 to 2.12	2.08 to 2.10	2.07 to 2.10
Zero strength time, seconds	176 to 300	301 to 750	115 to 175	176 to 300	301 to 750	751 to 2500	176 to 300	176 to 300
Deformation under load, at 250 pounds, 70°C. (158°F.), percent, max.	10	10	10	15	15	15	20	45
Melting point, °C.	210 to 220	210 to 220	200 to 210	200 to 210	200 to 210	200 to 210	195 to 205	180 to 190
°F.	(410 to 428)	(410 to 428)	(392 to 410)	(392 to 410)	(392 to 410)	(392 to 410)	(383 to 401)	(356 to 374)
Dielectric constant, max. at 1 MHz	2.50	2.50	2.50	2.50	2.50	2.50	2.50	2.50
Dissipation factor, max. at 1 MHz	0.012	0.012	0.012	0.012	0.012	0.012	0.015	0.015

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3.4 Color. The material (see 3.2) shall be natural, which is white in color and opaque, as determined by visual inspection. If another color is required by the procuring agency, the color and method of determination as well as any changes in property values (see 3.3) shall be as specified by the procuring agency (see 6.2)

3.5 Workmanship. The pellets, granules, or powder shall be uniform in color within each container and from container to container. The material shall be uniform in form; from container to container, and shall be free from contamination as determined by visual examination.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Sampling for inspection and acceptance. Sampling for inspection and acceptance shall be performed in accordance with the provisions set forth in MIL-STD-105, except where otherwise indicated. For purposes of sampling, an inspection lot for examination and tests shall consist of all material of the same type and class submitted for delivery at one time.

4.2.1 Inspection of materials of components. In accordance with 4.1, the supplier is responsible for insuring that materials and components used were manufactured, tested, and inspected in accordance with the requirements of referenced subsidiary specifications and standards to the extent specified, or if none, in accordance with the specification. In the event of conflict, this specification shall govern. A supplier's certificate of compliance with 3.1 shall be furnished.

4.2.2 Inspection of material.

4.2.2.1 Examination of the material. Examination of the material shall be made in accordance with the list of defects, inspection levels and acceptable quality levels (AQL's) set forth below. The lot size, for the purpose of determining the sample size in accordance with MIL-STD-105, shall be expressed in units of 100 pounds for examination in 4.2.2.1.1, and in units of shipping containers for examination in 4.2.2.1.2.

4.2.2.1.1 Examination of the material for defects in appearance and workmanship. The sample unit for this examination, specified in table II, shall be approximately one pound.

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TABLE II. Examination of the material for defects in appearance and workmanship

Examine	Defect
Appearance and workmanship	Form and color not uniform Form not as specified Color not as specified Not clean, presence of foreign material or other contamination

4.2.2.1.2 Examination of the preparation for delivery requirements. An examination shall be made in accordance with table III to determine that packing and marking comply with section 5 requirements. The sample unit for this examination shall be one shipping container fully packed, selected just prior to the closing operation. Shipping containers fully prepared for delivery shall be examined for closure defects.

TABLE III. Examination for preparation for delivery.

Examine	Defects
Packing	Not level specified; not in accordance with contract requirements. Any conconforming component, component missing, damaged or otherwise defective affecting serviceability. Inadequate application of components such as: Incomplete closures of case liners; container flaps, loose or inadequate strappings, bulged or distorted containers.
Quantity of material	Less than specified or indicated quantity.
Weight	Gross weight exceeds specified requirements.
Markings	Interior or exterior markings omitted, illegible, incorrect, incomplete, of improper size, location, sequence or method of application, or not in accordance with contract requirements.

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4.2.2.1.3 Inspection levels and acceptable quality levels (AQL's) for examinations. The inspection levels for determining the sample size and the acceptable quality level (AQL) expressed as defects per 100 units shall be as follows:

Examination paragraph	Inspection level	AQL
4.2.2.1.1	II	2.5
4.2.2.1.2	S-2	2.5

4.2.3 Testing. The material shall be tested for the characteristics listed in table I in accordance with the test methods specified herein. The lot size, for the purpose of determining the sample size for testing shall be expressed in units of 100 pounds. The sample unit shall consist of sufficient material to prepare all required specimens. The inspection level shall be S-1 with an acceptance number of 0. The results for each test shall be the averaged results of the specimens.

4.2.3.1 Classification of tests. All tests shall be classified as follows:

(a) Lot acceptance tests (4.2.3.2).
(b) Periodic lot check tests (4.2.3.3)

4.2.3.2 Lot acceptance tests. Lot acceptance tests shall be made on each lot of material and shall be the basis for acceptance or rejection of the lot, except when periodic lot check tests are required. Lot acceptance tests shall consist of tests for specific gravity, zero strength time, and melting point.

4.2.3.3 Periodic lot check tests. Periodic lot check tests shall be made on the first lot of material furnished under this specification, and on any subsequent lot specified by the procuring agency (see 6.2). Periodic lot check tests shall consist of tests for all properties specified in table I. When periodic lot check tests are made, they shall be included in the basis for acceptance or rejection of the lot.

4.3 Test methods.

4.3.1 Specimen preparation. Specimens shall be cut from compression molded sheet prepared in accordance with the method of specimen preparation specified in ASTM D 1430.

4.3.2 Specimen conditioning and testing. Specimens shall be conditioned in accordance with procedure A of ASTM D 618. Unless otherwise specified all testing shall be at 23° ± 2°C. (73.4° ± 3.6°F.) and 50 ± 5 percent relative humidity.

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4.3.3 Specific gravity. Three sepcimens shall be tested in accordance with method B of ASTM D 792 using n-heptane as the liquid.

4.3.4 Zero strength time. Two specimens shall be tested in accordance with ASTM D 1430.

4.3.5 Deformation under load. Three specimens shall be tested in accordance with method A of ASTM D 621 with the 250 pound (113.4 kilogram) load at 70° ± 1°C., except that the permissible deformations may exceed 25 percent for type III, class 2, grade B material only.

4.3.6 Melting point. Two specimens shall be tested in accordance with method D of ASTM D 2117. The test specimens shall be prepared by remolding a sample of the compression molded sheet specified in 4.3.1 to a film 0.4 to 1.6 mils thick between chrome plated metal plates in a press with platens heated to a surface temperature of 265° ± 5°C. (509° ± 9°F.).

4.3.7 Dielectric constant and dissipation factor. Three specimens shall be tested in accordance with ASTM D 150 using any electrode system specified in this test method. (A disc approximately 2 inches in diameter has been found satisfactory for this test.)

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging not required.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 Level A. Unless otherwide specified, the material shall be packed in one of the following types of containers:

(a) Fiber drums conforming to PPP-D-723, type II, grade A, or type III, grade A in quantities of 200 pounds maximum.
(b) Metal drums conforming to PPP-D-729, type III or type IV, in quantities of 400 pounds maximum.

Insofar as practical, containers shall be of uniform shape and size, with minimum cube and tare consistent with the protection required. Containers shall contain identical quantities and shall be closed in accordance with the applicable container specification. Fiber drums shall be furnished with an 0.004 inch thick polyethylene liner properly heat sealed.

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5.2.2 Level B. Unless otherwise specified, the material shall be in one of the following types of containers:

(a) Fiber drums conforming to PPP-D-723, type I, grade A in quantities of 200 pounds maximum.
(b) Metal drums conforming to PPP-D-729, type III or type IV, in quantities of 400 pounds maximum.

Insofar as practical, containers shall be of uniform shape and size with minimum cube and tare consistent with the protection required. Containers shall contain identical quantities and shall be closed in accordance with the applicable container specification. Fiber drums shall be furnished with an 0.004 inch thick polyethylene liner properly heat sealed.

5.2.3 Level C. Packing shall be in accordance with commercial practice adequate to insure acceptance and delivery by the carrier for the mode of transportation employed. Containers shall comply with the requirements of the Uniform Freight Classification Rules or National Motor Freight Classification Rules, as applicable to the mode of transportation.

5.3 Marking. In addition to any special marking required by the procurement document, marking of the exterior shipping containers shall be in accordance with Fed. Std. No. 123 for civil agencies or MIL-STD-129 for military agencies, as applicable (see 6.2).

6. NOTES

6.1 Intended use. Polychlorotrifluoroethylene plastics have a useful temperature range from approximately -253°C. (-423°F) to 199°C. (390°F.). In addition, the material exhibits good resistance to nearly all chemicals, a low coefficient of friction, low permeability to moisture and vapor, and good electrical properties. Applications include electrical and electronic components, corrosion resistant linings and barriers, insulation on wire and cable, and seals and gaskets for cryogenic equipment.

6.2 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Type, class and grade (where applicable) of material required (see 1.2)
(c) Form, if required (see 3.2).
(d) Particle size if required and method of determination (see 3.2).
(e) Color if required, and method of determination (see 3.4).
(f) Periodic lot check tests when required (see 4.2.3.3).
(g) Level of packing required (see 5.2).
(h) Special marking, when required (see 5.3)

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6.3 Determination of powder form. Resin shall be considered to be powder when apparent density is less than 0.800 grams per cubic centimeter when tested in accordance with ASTM D 1895 Apparent Density, Bulk Factor, and Pourability of Plastic Materials.

6.4 Physical and electrical properties. The physical and electrical properties of parts molded or extruded from polychlorotrifluoroethylene plastics vary with the final crystalline content which is determined by processing and subsequent annealing conditions. Accordingly, the property values listed in table I apply only to the test specimens prepared as specified in this document. These values may not be applicable as design criteria to parts prepared and annealed under other conditions.

6.5 Cross index. Table IV shows the classification of material covered by this specification and the corresponding classification of material specified in previous issues of L-P-385A and MIL-M-21470(NORD).

TABLE IV. Classification in this document and previous issues of corresponding classification in L-P-385 and MIL-M-21470(NORD)

Classification in documents L-P-385B and L-P-385C	Classification in L-P-385A	Classification in MIL-M-21470(NORD)
Type I	-	-
Class 2	Grade 2	Grade 2
Class 3	Grade 3	Grade 3
Type II	-	-
Class 1	-	Grade 1
Class 2	-	-
Class 3	-	-
Class 4	-	Grade 4
Type III	-	-
Class 2	-	-
Grade A	-	-
Grade B	-	-

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